SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

I jeshyapriya.R(192524453) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

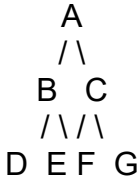
jeshyapriya.R(192524453)

Annexure A

Dry Run Evaluation

QUESTION 1: Breadth-First Search (BFS)

Given Graph:



BFS Algorithm:
Breadth-First Search explores nodes level by level using a queue (FIFO).

Step-by-Step Table:

Iteration	Node Visited	Queue After Iteration	Visited Nodes
1	A	B, C	A
2	B	C, D, E	A, B
3	C	D, E, F, G	A, B, C
4	D	E, F, G	A, B, C, D
5	E	F, G	A, B, C, D, E
6	F	G	A, B, C, D, E, F
7	G	Empty	A, B, C, D, E, F, G

1.BFS Traversal Order:
A → B → C → D → E → F → G

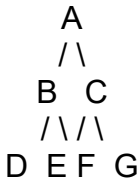
2. Queue Contents After Each Iteration:

1. After visiting A: [B, C]
2. After visiting B: [C, D, E]
3. After visiting C: [D, E, F, G]
4. After visiting D: [E, F, G]
5. After visiting E: [F, G]
6. After visiting F: [G]
7. After visiting G: []

Result: BFS successfully visits all nodes level by level.

QUESTION 2: Depth-First Search (DFS)

Given Graph:



DFS Algorithm:
Depth-First Search explores one branch as deeply as possible before backtracking. It can be implemented using a stack or recursion.
Assume the left child is visited before the right child.

Step-by-Step Traversal:

Iteration	Node Visited	Explanation
1	A	Start from the root node
2	B	Move to the left child
3	D	Move to the left child of B
4	E	Backtrack to B and visit E
5	C	Backtrack to A and visit C
6	F	Visit the left child of C
7	G	Visit the right child of C

DFS Traversal Order:

A → B → D → E → C → F → G
Result: DFS successfully visits all nodes by exploring each branch before backtracking.

QUESTION 3: Breadth-First Search (BFS) for the 8-Puzzle Problem

Initial State:
1 3 6
5 0 2
4 7 8
Goal State:
1 2 3
4 5 6
7 8 0
Here, 0 represents the empty space.

BFS Algorithm:
BFS explores states level by level using a queue. It generates possible states by moving the empty space up, down, left, or right, whenever the move is valid.

1. Solution Path
The following is a shortest solution path from the initial state to the goal state.
Initial State:

1 3 6
5 0 2
4 7 8

Step 1 – Move empty space Right:

1 3 6
5 2 0
4 7 8

Step 2 – Move empty space Up:

1 3 0
5 2 6
4 7 8

Step 3 – Move empty space Left:

1 0 3
5 2 6
4 7 8

Step 4 – Move empty space Down:

1 2 3
5 0 6
4 7 8

Step 5 – Move empty space Left:

1 2 3
0 5 6
4 7 8

Step 6 – Move empty space Down:

1 2 3
4 5 6
0 7 8

Step 7 – Move empty space Right:

1 2 3
4 5 6
7 0 8

Step 8 – Move empty space Right:

1 2 3
4 5 6
7 8 0

The goal state is reached after 8 moves.

2. BFS Dry Run Table

The first five iterations of the BFS search are shown below. The queue contains states waiting to be expanded. For readability, the queue is summarized rather than listing every state.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	Initial state	Initial state	4 new states
2	State with empty space at position (1,2)	3 states	5 states
3	State with empty space at position (2,1)	4 states	6 states
4	State with empty space at position (2,0)	5 states	7 states
5	State with empty space at position (1,2)	6 states	8 states

3. Which Node Is Expanded in Each Iteration?

BFS expands the oldest state in the queue first. It starts with the initial state and continues expanding states in the order they were discovered until the goal state is found.

4. How Many States Are Generated in Total?

The solution path contains 9 states, including the initial and goal states, and requires 8 moves.
The total number of states explored by BFS can be larger than 9 because BFS also examines alternative states. The exact count depends on the order in which valid moves are generated.

5. Why Does BFS Always Find the Shortest Solution in an Unweighted 8-Puzzle Problem?

BFS explores all states at one depth before moving to the next depth. Therefore, it finds the goal state using the minimum number of moves when each move has the same cost.

Result: The 8-puzzle goal state is reached in 8 moves using BFS, which guarantees a shortest solution.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.